

1. What is a semiconductor? What is an extrinsic semiconductor? Describe the difference between P-type and N-type semiconductor materials.
2. At what temperature are there no electrons in the conduction band of a semiconductor?
3. As one electron is promoted from the valence band to the conduction band, a _____ is formed in the valence band.
4. Group _____ elements are used as dopants to produce n-type semiconductors, because they have _____ than the original Group IV material.
5. Group _____ elements are used as dopants to produce p-type semiconductors, because they have _____ than the original Group IV material.
6. A diode contains both _____ and _____ regions.
7. For current to flow through a diode, the positive terminal of the power supply must be connected to the _____-type material.
8. Majority of integrated circuits utilize a Si (100) substrate, Why?

Chapter IV

1. Semiconductor definition

According to the conductivity between conductors and insulators.

2. Intrinsic and extrinsic semiconductors

- Intrinsic: $N_{\text{electron}} = N_{\text{hole}}$
- Extrinsic $N_{\text{electron}} \neq N_{\text{hole}}$
- n-type and p-type doping

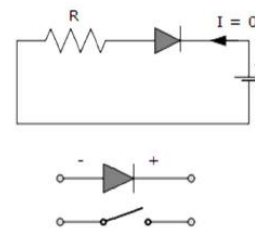
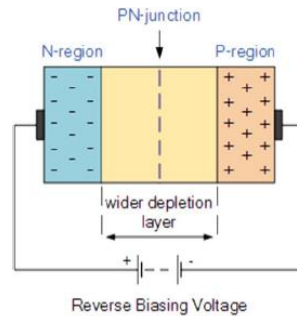
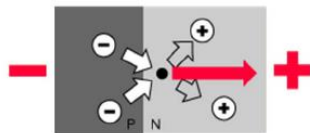
3. Silicon wafer production

- Silicon wafer is fabricated from sea sand.
- Metallurgical-grade silicon (>98%) and electronic-grade silicon (99.999999999%).
- Crystallographic planes of Si: (100) (110) (111); Why do the majority of ICs utilize the Si (100) orientation?

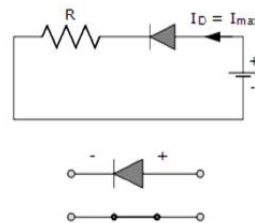
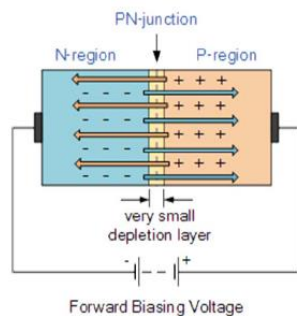
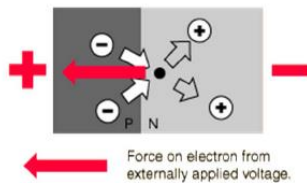
4. P-N junction diode

- Depleted region
- Reverse bias and forward bias

Reverse Bias



Forward Bias



5. MOSFETs (metal oxide semiconductor field effect transistors)

- NMOSFET n-channel with p-type substrate
- PMOSFET p-channel with n-type substrate
- Threshold voltage

- CMOS (complementary metal oxide semiconductor)

6. IC fabrication (you are required to briefly know the process)